WORKED SOLUTION

STEP I 1991, Question 15

Probability and Statistics • Normal Approximation

STEP I 1991, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q15

The key idea

Each X_j has mean 0 and variance 1, and the throws are independent, so S_n has mean 0 and variance n; for large n the central limit theorem lets us treat S_n as approximately normal. The final part looks similar but has a trap: every Y_k contains the *same* first throw X_1, so the Y_k are not independent, and that one shared throw ends up dominating the whole total.

Solution

Each X_j takes the values ± 1 with probability $\frac{1}{2}$ each, so

$$\mathrm{E}(X_j) = 0, \qquad \mathrm{E}(X_j^2) = 1, \qquad \mathrm{Var}(X_j) = 1.$$

The throws are independent, so $\mathrm{E}(S_n) = 0$ and $\mathrm{Var}(S_n) = n$. By the central limit theorem, for large n,

$$S_n \approx \mathrm{N}(0,\, n).$$

(Equivalently, if H is the number of heads then $H \sim \mathrm{B}(n, \frac{1}{2})$ and $S_n = H - (n - H) = 2H - n$.)

Part (i)

With $n = 10\,000$ the standard deviation of S_n is $\sqrt{10\,000} = 100$, so

$$\mathrm{P}(S_n > 100) \approx \mathrm{P}\left(Z > \frac{100}{100}\right) = 1 - \Phi(1) \approx 0.16.$$

If you want to be a little more careful, use a continuity correction on H: $S_n > 100$ means $2H - 10\,000 > 100$, that is $H \geq 5051$. Since H has mean 5000 and standard deviation 50,

$$\mathrm{P}(H \geq 5051) \approx \mathrm{P}\left(Z > \frac{5050.5 - 5000}{50}\right) = 1 - \Phi(1.01) \approx 0.156.$$

Either way the answer is about 0.16.

Part (ii)

Standardising,

$$\mathrm{P}(S_n > 0.01n) \approx \mathrm{P}\left(Z > \frac{0.01n}{\sqrt{n}}\right) = 1 - \Phi\left(0.01\sqrt{n}\right).$$

This decreases as n increases, and it is less than 0.01 exactly when $\Phi(0.01\sqrt{n}) > 0.99$. From tables, $\Phi(2.326) = 0.99$, so we need

$$0.01\sqrt{n} > 2.326 \quad \Longleftrightarrow \quad \sqrt{n} > 232.6 \quad \Longleftrightarrow \quad n > 54\,100 \text{ (approximately).}$$

So the least such n is roughly 54 000. (Using 2.33 instead gives about 54 300; at this level of approximation any answer close to 5.4×10^4 is the right one.)

The new scoring

For $2 \le k \le n$ there are four cases for the first throw and the kth throw:

first throw	kth throw	score Y_k	$X_1 + X_k$
H	H	2	$1+1=2$
T	T	-2	$-1-1=-2$
H	T	0	$1-1=0$
T	H	0	$-1+1=0$

The last two columns agree in every case, so $Y_k = X_1 + X_k$, as required.

Mean and variance. By linearity, $\mathrm{E}(Y_k) = \mathrm{E}(X_1) + \mathrm{E}(X_k) = 0$. Since $k \ge 2$, the throws X_1 and X_k are different throws and so independent, giving

$$\mathrm{Var}(Y_k) = \mathrm{Var}(X_1) + \mathrm{Var}(X_k) = 2.$$

(Directly: Y_k is 2, -2, 0 with probabilities $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$, so $\mathrm{E}(Y_k^2) = 2$.)

The limit. Write $m = n - 1$ and $T = Y_2 + Y_3 + \cdots + Y_n$. Then

$$T = \sum_{k=2}^{n} (X_1 + X_k) = mX_1 + S', \qquad \text{where } S' = X_2 + X_3 + \cdots + X_n.$$

Ask yourself Each Y_k has mean 0 and variance 2, just like a scaled X_j. Why can't we simply copy the argument of part (ii)?

Because the argument of part (ii) needed the terms to be independent, and the Y_k are not: they all share X_1. Indeed for $j \ne k$, $\mathrm{Cov}(Y_j, Y_k) = \mathrm{Var}(X_1) = 1$, and $\mathrm{Var}(T) = m^2 + m$, not $2m$. So we look at the two values of X_1 separately. Note that S' is a sum of m terms each ± 1, so $-m \le S' \le m$.

- If the first throw is a tail, $X_1 = -1$ and $T = -m + S' \leq 0$. So $T > 0.01m$ is impossible.
- If the first throw is a head, $X_1 = 1$ and $T = m + S'$, so $T > 0.01m$ exactly when $S' > -0.99m$. By the same normal approximation as in part (ii), with $S' \approx \mathrm{N}(0, m)$,

$$\mathrm{P}(S' \leq -0.99m) \approx \Phi\left(-0.99\sqrt{m}\right) \to 0 \quad \text{as } m \to \infty.$$

  (No approximation is needed: by Chebyshev's inequality this probability is at most $m/(0.99m)^2 \to 0$.)

Since S' is independent of X_1,

$$\mathrm{P}(T > 0.01m) = \tfrac{1}{2} \cdot 0 + \tfrac{1}{2}\,\mathrm{P}(S' > -0.99m) \;\to\; \tfrac{1}{2} \quad \text{as } n \to \infty.$$

So the statement is **false**: the probability tends to $\frac{1}{2}$, not 0. In words, for large n the total is positive essentially when the first throw is a head, and otherwise it is not.

Checks

- Part (i): the exact binomial probability $\mathrm{P}(H \geq 5051)$ for $H \sim \mathrm{B}(10\,000, \frac{1}{2})$ is 0.15625, matching the continuity-corrected value 0.156.
- Part (ii): the exact binomial value of $\mathrm{P}(S_n > 0.01n)$ is 0.0125 at $n = 50\,000$, about 0.0100 near $n = 54\,000$, and 0.0094 at $n = 55\,000$, so the threshold is indeed close to 54 000.
- Final part: enumerating all 2^n outcomes for $n \leq 14$ agrees exactly with $\frac{1}{2}\,\mathrm{P}(S' > -0.99m)$; at $n = 14$ the probability is already 0.49994, and a simulation with $n = 100$ gave 0.497.

Common slips

- Treating $Y_2, \ldots, Y_n$ as independent and concluding $\mathrm{Var}(T) = 2(n-1)$, which makes the probability appear to tend to 0.
- Using $\mathrm{Var}(S_n) = n/4$ (the variance of the number of heads) instead of n for the score.

Beyond STEP

Part (ii) is a quantitative form of the weak law of large numbers: S_n/n is very unlikely to be far from 0 once n is large. The final part shows how that law fails when the terms are strongly dependent: here every term shares the first throw, so the average $T/m = X_1 + S'/m$ tends to X_1, a random quantity, rather than to the mean 0. Laws of large numbers for dependent sequences (ergodic theorems, exchangeable sequences) are a rich topic in undergraduate probability; Grimmett and Welsh, *Probability: An Introduction*, covers the weak law and Chebyshev's inequality.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.